Multimodal RAG Architectures: An Empirical Benchmark

AI Research Labs — Technical Report #2025-04

Model Pipeline	Table QA Accuracy	Chart Reasoning	Hallucination Rate
Text-Only RAG	62.4%	14.2%	34.8%
OCR + Text RAG	74.1%	28.5%	22.6%
DocuLens Multimodal	92.8%	89.4%	4.1%

Abstract.
We evaluate multimodal document intelligence frameworks across complex documents containing dense tabular structures, high-resolution chart images, and multi-column typography. Our findings indicate that multimodal grounding achieves a 94.2% factual precision rate, drastically reducing hallucinations compared to text-only baselines.